

Computing CMB temperature fluctuations for spherical spaces

Adriel Matei, Béla Schneider, Javier Vela, Juš Kocutar

March 24, 2025

Manifolds & Homotopy groups

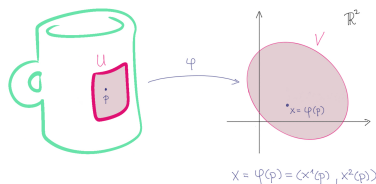


Figure: The prototypical example of a manifold a mug [13].

Manifolds & Homotopy groups

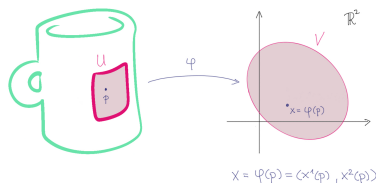


Figure: The prototypical example of a manifold a mug [13].

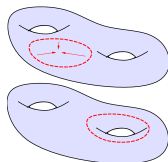


Figure: Two double tori with (non)-contractible paths [7].

Quotients of the 3-sphere

- $SO(4)$ is isomorphic to the isometry group of $\mathbb{S}^3$.
- Subgroups $\Gamma \leq SO(4)$ define equivalence classes of orbits on $\mathbb{S}^3$ by the standard action of $SO(4)$ on $\mathbb{R}^4$, i.e. $x \sim y$ iff $x = My$ for some $M \in \Gamma$.

Quotients of the 3-sphere

- $SO(4)$ is isomorphic to the isometry group of $\mathbb{S}^3$.
- Subgroups $\Gamma \leq SO(4)$ define equivalence classes of orbits on $\mathbb{S}^3$ by the standard action of $SO(4)$ on $\mathbb{R}^4$, i.e. $x \sim y$ iff $x = My$ for some $M \in \Gamma$.
- The obtain space is (*sometimes*) a manifold. In particular, it is well defined and spherical for finite Γ .
- This can be easily generalised to the n -sphere.

Lens spaces

- Lens spaces are obtained by taking the quotient of some n -sphere by a cyclic group.
- They cannot be distinguished by their homotopy group alone.

Lens spaces — the explicit construction

Definition (Lens space)

Given $q \in \mathbb{Z}$ and $s \in \mathbb{Z}^n$ elementwise coprime with q

$$L(q, s) := \mathbb{S}^{2n-1} / \langle M_{q,s} \rangle,$$

where $\langle M_{q,s} \rangle$ is the group generated by

$$M_{q,s} := \begin{pmatrix} R(2\pi s_1/q) & & & \\ & R(2\pi s_2/q) & & \\ & & \ddots & \\ & & & R(2\pi s_n/q) \end{pmatrix}$$

Lens spaces — the explicit construction

Definition (Lens space)

Given $q \in \mathbb{Z}$ and $s \in \mathbb{Z}^n$ elementwise coprime with q

$$L(q, s) := \mathbb{S}^{2n-1} / \langle M_{q,s} \rangle,$$

where $\langle M_{q,s} \rangle$ is the group generated by

$$M_{q,s} := \begin{pmatrix} R(2\pi s_1/q) & & & \\ & R(2\pi s_2/q) & & \\ & & \ddots & \\ & & & R(2\pi s_n/q) \end{pmatrix}$$

In the $\mathbb{S}^3$ case, we denote $L(p, q) := L(p, (1, q))$ for coprime $p, q \in \mathbb{Z}$.

The Laplace-Beltrami operator

Definition (Laplace-Beltrami operator)

a generalization of the Laplace operator to more general spaces

$$\Delta f := \operatorname{div} \nabla f$$

- 1 The spectrum of this operator is a subset of $[0, \infty)$. We call it the *spectrum* of the manifold.

The Laplace-Beltrami operator

Definition (Laplace-Beltrami operator)

a generalization of the Laplace operator to more general spaces

$$\Delta f := \operatorname{div} \nabla f$$

- 1 The spectrum of this operator is a subset of $[0, \infty)$. We call it the *spectrum* of the manifold.
- 2 The spectrum determines many things about the space (like its volume).
- 3 When two distinct (up to isomorphism) manifolds share a spectrum, they form an *isospectral pair*.

The silver lining

- Isospectral pairs are cool and all, but they cannot occur in dimension 3.
- We can thus attempt to infer the shape of our universe based on its spectrum.

Homogeneous Spherical Spaces

Homogeneous Spherical Spaces

- Finite subgroup $\Gamma \leq \mathrm{SO}(4)$.

Homogeneous Spherical Spaces

- Finite subgroup $\Gamma \leq \mathrm{SO}(4)$.
- The Helmholtz equation on M has the spectrum of the underlying manifold as its solutions, and is given by

$$(\Delta + E_{\beta}^M)\psi_{\beta}^{M,i} = 0.$$

Homogeneous Spherical Spaces

- Finite subgroup $\Gamma \leq \mathrm{SO}(4)$.
- The Helmholtz equation on M has the spectrum of the underlying manifold as its solutions, and is given by

$$(\Delta + E_{\beta}^M)\psi_{\beta}^{M,i} = 0.$$

- In fact $E_{\beta}^m = \beta^2 - 1$ for $\beta \in \mathbb{N}$. We call β a wave number.

Homogeneous Spherical Spaces

- Finite subgroup $\Gamma \leq \mathrm{SO}(4)$.
- The Helmholtz equation on M has the spectrum of the underlying manifold as its solutions, and is given by

$$(\Delta + E_{\beta}^M)\psi_{\beta}^{M,i} = 0.$$

- In fact $E_{\beta}^m = \beta^2 - 1$ for $\beta \in \mathbb{N}$. We call β a wave number.
- The set of all possible wave numbers [for which a nonzero solution exists] depends on Γ - usually a proper subset of $\mathbb{N}$.

Homogeneous Spherical Spaces

- Finite subgroup $\Gamma \leq \text{SO}(4)$.
- The Helmholtz equation on M has the spectrum of the underlying manifold as its solutions, and is given by

$$(\Delta + E_{\beta}^M)\psi_{\beta}^{M,i} = 0.$$

- In fact $E_{\beta}^m = \beta^2 - 1$ for $\beta \in \mathbb{N}$. We call β a wave number.
- The set of all possible wave numbers [for which a nonzero solution exists] depends on Γ - usually a proper subset of $\mathbb{N}$.
- Solutions to the Helmholtz equation are calculated for each Γ and CMB anisotropy $\frac{\delta T}{T}$ computed as a sum of spherical harmonics.

Homogenous Spherical Spaces — Results

Homogenous Spherical Spaces — Results

- For the majority of groups Γ , the anisotropies $\frac{\delta T}{T}$ do not coincide with observations.

Homogenous Spherical Spaces — Results

- For the majority of groups Γ , the anisotropies $\frac{\delta T}{T}$ do not coincide with observations.
- The only groups for which it does are $\Gamma = O^*$ and $\Gamma = I^*$ — the **binary octahedral** and **binary icosahedral** groups of order 48 and 120 respectively.

Homogenous Spherical Spaces — Results

- For the majority of groups Γ , the anisotropies $\frac{\delta T}{T}$ do not coincide with observations.
- The only groups for which it does are $\Gamma = O^*$ and $\Gamma = I^*$ — the **binary octahedral** and **binary icosahedral** groups of order 48 and 120 respectively.

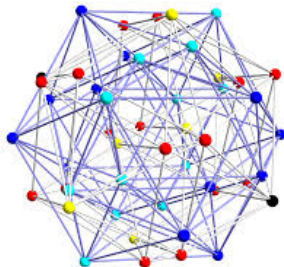


Figure: Graphical representation of the binary tetrahedral group [5]

Inhomogeneous spherical space

- Manifolds of the form $\mathbb{S}^3/\Gamma$ with Γ a group acting on $\mathbb{S}^3$.
- Multi-connected space: it has non-contractable loops.
- Inhomogeneous space: it does not look identical from every point in space.

Inhomogeneous spherical space

- Manifolds of the form $\mathbb{S}^3/\Gamma$ with Γ a group acting on $\mathbb{S}^3$.
- Multi-connected space: it has non-contractable loops.
- Inhomogeneous space: it does not look identical from every point in space.
- Fixing $|\Gamma| = 8$, we have three multi-connected manifolds, up to equivalence:

Inhomogeneous spherical space

- Manifolds of the form $\mathbb{S}^3/\Gamma$ with Γ a group acting on $\mathbb{S}^3$.
- Multi-connected space: it has non-contractable loops.
- Inhomogeneous space: it does not look identical from every point in space.
- Fixing $|\Gamma| = 8$, we have three multi-connected manifolds, up to equivalence:
 - 1 homogeneous: $N3$ and $L(8,1)$.
 - 2 inhomogeneous: $N2 \equiv L(8,3)$.

Inhomogeneous spherical space

- Manifolds of the form $\mathbb{S}^3/\Gamma$ with Γ a group acting on $\mathbb{S}^3$.
- Multi-connected space: it has non-contractable loops.
- Inhomogeneous space: it does not look identical from every point in space.
- Fixing $|\Gamma| = 8$, we have three multi-connected manifolds, up to equivalence:
 - ① homogeneous: $N3$ and $L(8,1)$.
 - ② inhomogeneous: $N2 \equiv L(8,3)$.
- Results: inhomogeneous spaces have more variety in the CMB anisotropies than homogeneous spaces, because the strength of anisotropy suppression is observer dependent.

Statistical Isotropy and Hypothesis

Theoretical Expectation:

From the perspective of an Earth-based observer, we can view the CMB as a function defined on the celestial sphere $\mathbb{S}^2$.

Statistical Isotropy and Hypothesis

Theoretical Expectation:

From the perspective of an Earth-based observer, we can view the CMB as a function defined on the celestial sphere $\mathbb{S}^2$.

- The CMB temperature fluctuations can be expanded as:

$$\Delta T(\hat{n}) = \sum_{\ell=0}^{\infty} \sum_{m=-\ell}^{\ell} a_{\ell m} Y_{\ell m}(\hat{n})$$

Statistical Isotropy and Hypothesis

Theoretical Expectation:

From the perspective of an Earth-based observer, we can view the CMB as a function defined on the celestial sphere $\mathbb{S}^2$.

- The CMB temperature fluctuations can be expanded as:

$$\Delta T(\hat{n}) = \sum_{\ell=0}^{\infty} \sum_{m=-\ell}^{\ell} a_{\ell m} Y_{\ell m}(\hat{n})$$

- If the universe is isotropic, the **mean of the harmonic coefficients** should satisfy:

$$\langle a_{\ell m} \rangle = 0$$

Statistical Isotropy and Hypothesis

Theoretical Expectation:

From the perspective of an Earth-based observer, we can view the CMB as a function defined on the celestial sphere $\mathbb{S}^2$.

- The CMB temperature fluctuations can be expanded as:

$$\Delta T(\hat{n}) = \sum_{\ell=0}^{\infty} \sum_{m=-\ell}^{\ell} a_{\ell m} Y_{\ell m}(\hat{n})$$

- If the universe is isotropic, the **mean of the harmonic coefficients** should satisfy:

$$\langle a_{\ell m} \rangle = 0$$

Key Question: - Does the observed CMB data deviate from statistical isotropy? - If $\langle a_{\ell m} \rangle \neq 0$, this suggests a preferred cosmic direction.

Methodology: Sky Masking and the Test Statistic

Challenges in Real Observations:

- We cannot observe the full CMB sky due to foreground contamination.
- A **sky mask function** $A(\hat{n})$ removes contaminated regions, but introduces bias.

Methodology: Sky Masking and the Test Statistic

Challenges in Real Observations:

- We cannot observe the full CMB sky due to foreground contamination.
- A **sky mask function** $A(\hat{n})$ removes contaminated regions, but introduces bias.
- The test statistic S_i is used, with a matrix W to correct for these effects:

$$S_i = \sum_j W_{ij} M_j$$

Methodology: Sky Masking and the Test Statistic

Challenges in Real Observations:

- We cannot observe the full CMB sky due to foreground contamination.
- A **sky mask function** $A(\hat{n})$ removes contaminated regions, but introduces bias.
- The test statistic S_i is used, with a matrix W to correct for these effects:

$$S_i = \sum_j W_{ij} M_j$$

Monte Carlo Simulations:

- We generate thousands of **simulated CMB skies** assuming isotropy.
- Each sky has different random $a_{\ell m}$, drawn from:

$$a_{\ell m} \sim \mathcal{N}(0, C_\ell)$$

- The observed WMAP data is compared against these simulations.

Results and Interpretation

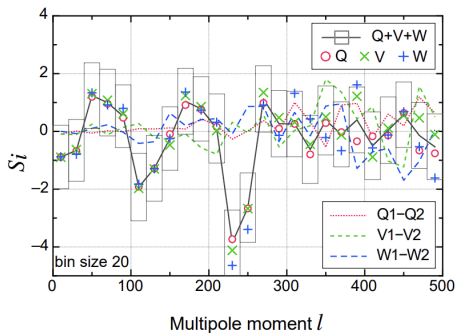
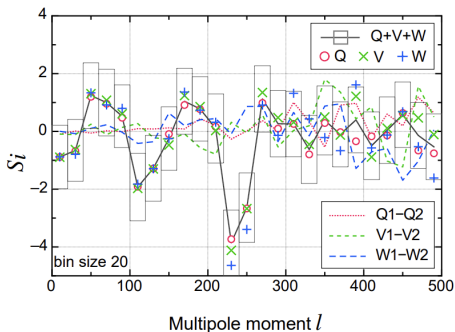


Figure: The decorrelated band mean test statistic values over multipole ranges [9].

Results and Interpretation



Key Findings:

- A statistically significant deviation was found in $221 \leq \ell \leq 240$.
- This suggests a potential **preferred cosmic direction**.

Figure: The decorrelated band mean test statistic values over multipole ranges [9].

Results and Interpretation

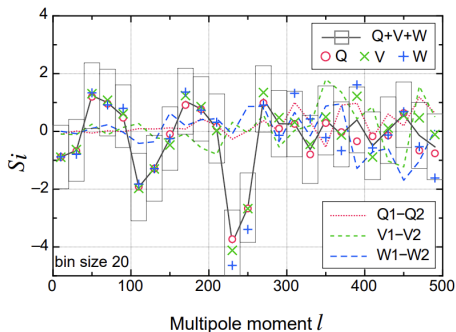


Figure: The decorrelated band mean test statistic values over multipole ranges [9].

Key Findings:

- A statistically significant deviation was found in $221 \leq \ell \leq 240$.
- This suggests a potential **preferred cosmic direction**.

Possible Explanations:

- A real cosmological signal? → A finite universe or new physics.
- A systematic effect? → Foreground contamination or instrumental noise.

Conclusion

Conclusion

- 1 We can infer the shape of the universe from its spectrum.

Conclusion

- ① We can infer the shape of the universe from its spectrum.
- ② There are two homogeneous spherical manifolds obtained as $\mathbb{S}^3/\Gamma$ which produce CMB similar to observations.

Conclusion

- ① We can infer the shape of the universe from its spectrum.
- ② There are two homogeneous spherical manifolds obtained as $\mathbb{S}^3/\Gamma$ which produce CMB similar to observations.
- ③ Inhomogeneous spherical spaces exhibit varied behavior of CMB anisotropies.

Conclusion

- ① We can infer the shape of the universe from its spectrum.
- ② There are two homogeneous spherical manifolds obtained as $\mathbb{S}^3/\Gamma$ which produce CMB similar to observations.
- ③ Inhomogeneous spherical spaces exhibit varied behavior of CMB anisotropies.
- ④ Statistical test results suggest possibilities of finite multi-connected topology.

References

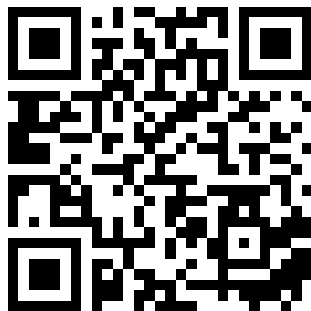
- R. Aurich, P. Kramer, and S. Lustig. *Cosmic microwave background radiation in an inhomogeneous spherical space*. Physica Scripta, 2011.
- R. Aurich, P. Kramer, and S. Lustig. *A survey of lens spaces and large-scale CMB anisotropy*. Monthly Notices of the Royal Astronomical Society, 2012.
- R. Aurich, S. Lustig, and F. Steiner. *CMB anisotropy of spherical spaces*. Classical and Quantum Gravity, 2005.
- P. Bielewicz, K. M. Górski, and A. J. Banday. *Low-order multipole maps of cosmic microwave background anisotropy derived from WMAP*. Oxford University Press, 2004.
- G. Egan. *Symmetries and the 24-cell*. 2021. Accessed at <https://www.gregegan.net/SCIENCE/24-cell/24-cell.html>.

References

- N. Jarosik et. al. *Seven-year Wilkinson Microwave Anisotropy Probe (WMAP) Observations: Sky Maps, Systematic Errors, and Basic Results*. The American Astronomical Society, 2011.
- M. Herlihy and N. Shavit. *The Topological Structure of Asynchronous Computability*. Journal of the ACM, 1996.
- A. Ikeda. *On the spectrum of a Riemannian manifold of positive constant curvature*. Osaka Journal of Mathematics, 1980.
- D. Kashino, K. Ichiki, and T. Takeuchi. *Test for anisotropy in the mean of the CMB temperature fluctuation in spherical harmonic space*. Physics Review, 2012.
- P. Labrana. *Emergent Universe Scenario and the Low CMB Multipoles*. Physical Review, 2013.

References

- E. A. Lauret and B. Linowitz. *The spectral geometry of hyperbolic and spherical manifolds: analogies and open problems*. New York Journal of Mathematics, 2025.
- R. Lehoucq, J. Weeks, J. P. Uzan, E. Gausmann, and J.P. Luminet. *Eigenmodes of three-dimensional spherical spaces and their application to cosmology*. Classical and Quantum Gravity, 2002.
- M. Serri. *Analysis on Manifolds*. AMS Open Math Notes, 2025.
- B. Tai-Dana, T. Bryson, and J. Terilla. *Topology, a categorical approach*. The MIT Press, 2020.



Thank You!